$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

$f_n(x) \to f(x)$

$e^{ix} = \cos x + i \sin x$

Chapter 7: Sequences and Series of Functions

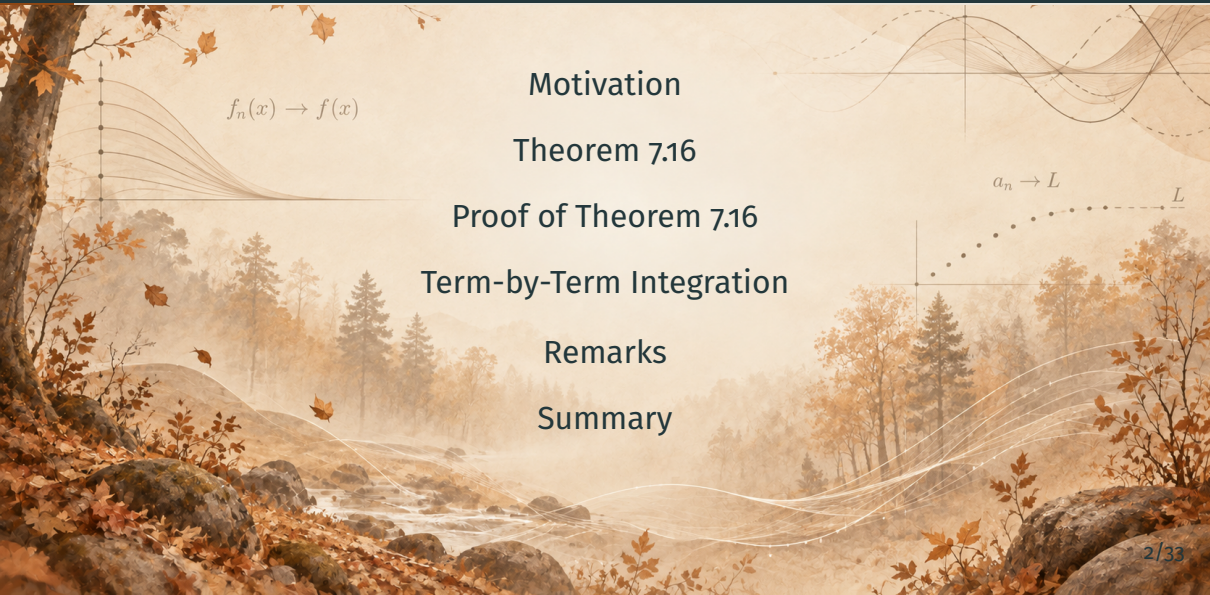
Section: Uniform Convergence and Integration

$a_n \to L$

Based on Walter Rudin's *Principles of Mathematical Analysis*

$S_N = \sum_{n=1}^{N} u_n$

Outline



Motivation

Theorem 7.16

Proof of Theorem 7.16

Term-by-Term Integration

Remarks

Summary



Motivation

Why This Section Matters

- The main problem of Section 7.1, now for integrals:

$$\lim_{n\to\infty} \int_a^b f_n \, d\alpha \stackrel{?}{=} \int_a^b \left(\lim_{n\to\infty} f_n \right) d\alpha$$

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- Under *pointwise* convergence: no.
  Example 7.6: integrals tend to $\frac{1}{2}$, the limit function is 0

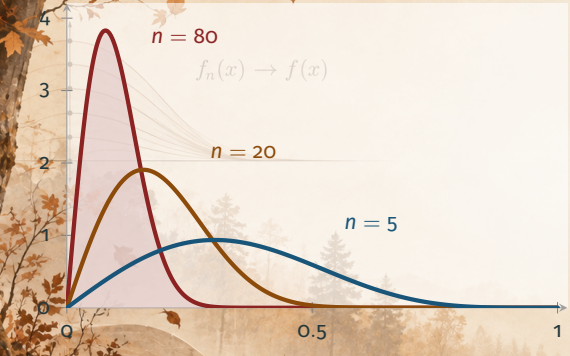
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- Under *pointwise* convergence: no.
  Example 7.6: integrals tend to $\frac{1}{2}$, the limit function is 0
- Under *uniform* convergence on $[a, b]$: yes!

Recall Example 7.6: Bumps That Do Not Vanish



Example 7.6 (with n)

$$f_n(x) = n\,x(1-x^2)^n \text{ on } [0,1]$$

$$f_n(x) \to 0 \quad (0 \le x \le 1)$$

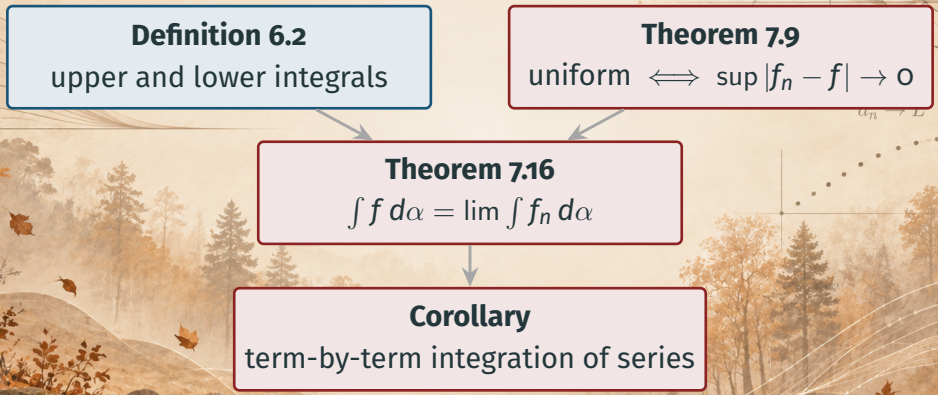
$$\int_0^1 f_n\,dx = \frac{n}{2n+2} \to \frac{1}{2}$$

Rudin's first version

With n^2 in place of n:

$$\int_0^1 f_n\,dx = \frac{n^2}{2n+2} \to +\infty$$

Roadmap of This Section





Theorem 7.16

Theorem 7.16: Limit of Integrals

Theorem 7.16

Let α be monotonically increasing on $[a, b]$. Suppose $f_n \in \mathscr{R}(\alpha)$ on $[a, b]$, for $n = 1, 2, 3, \ldots$, and suppose $f_n \to f$ uniformly on $[a, b]$. Then $f \in \mathscr{R}(\alpha)$ on $[a, b]$, and

$$\int_a^b f \, d\alpha = \lim_{n \to \infty} \int_a^b f_n \, d\alpha. \tag{23}$$

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Three conclusions in one

(i) f is integrable **(ii)** the limit on the right exists **(iii)** they are equal

Why Integrability of f Is Not Free

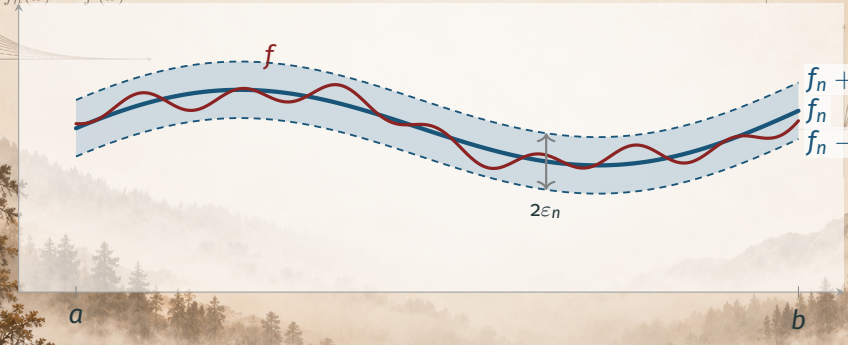
Remark

- If the f_n are *continuous*: f is continuous (Theorem 7.12), hence $f \in \mathscr{R}(\alpha)$ (Theorem 6.8). Easy.
- But $f_n \in \mathscr{R}(\alpha)$ may be discontinuous. Example 7.4: each f_m is nonzero at finitely many points of $[0, 1]$, so $f_m \in \mathscr{R}$, yet

$$\lim_{m \to \infty} f_m(x) = \begin{cases} 1 & x \text{ rational} \\ 0 & x \text{ irrational} \end{cases}$$

is not Riemann-integrable.

The Picture: f Trapped in a Thin Tube



$S_N = \sum_{n=1}^{N} u_n$

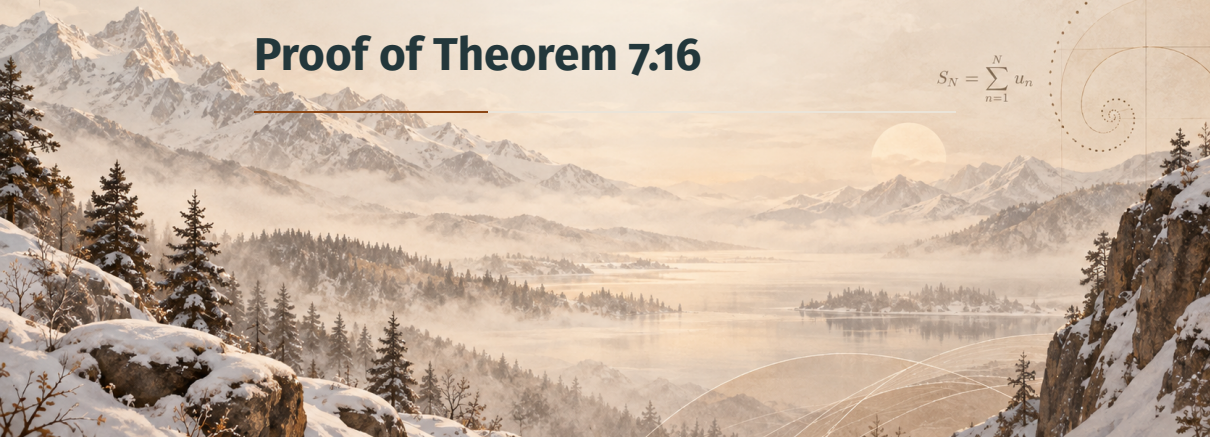
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Proof of Theorem 7.16

$S_N = \sum_{n=1}^{N} u_n$



Proof Strategy: Theorem 7.16

Goal: $f \in \mathscr{R}(\alpha)$ and $\int_a^b f_n \, d\alpha \to \int_a^b f \, d\alpha$.

Approach:

1. Trap f in a tube: $f_n - \varepsilon_n \leq f \leq f_n + \varepsilon_n$.

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2. Sandwich the upper and lower integrals of f; their gap $\to 0$, so $f \in \mathscr{R}(\alpha)$.
3. The same sandwich gives $\left| \int f \, d\alpha - \int f_n \, d\alpha \right| \le \varepsilon_n [\alpha(b) - \alpha(a)] \to 0$.

Tools We Need

Recall: Definition 6.2 (upper and lower integrals)

For bounded f on $[a, b]$:

$$\overline{\int_a^b} f \, d\alpha = \inf_P U(P, f, \alpha), \qquad \underline{\int_a^b} f \, d\alpha = \sup_P L(P, f, \alpha).$$

$f \in \mathscr{R}(\alpha) \iff$ the two are equal.

Consequence: $g \leq h \Rightarrow$ upper/lower integrals of $g \leq$ those of h.

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Recall: Theorem 7.9

$f_n \to f$ uniformly on $E \iff M_n = \sup_{x \in E} |f_n(x) - f(x)| \to 0$.

Proof of Theorem 7.16

Step 1: Reduce to real f_n; measure the error

Real and imaginary parts are treated separately, so assume f_n real. Put

$$\varepsilon_n = \sup_{a \le x \le b} |f_n(x) - f(x)|. \tag{24}$$

By Theorem 7.9, $\varepsilon_n \to 0$ as $n \to \infty$.

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By Theorem 7.9, $\varepsilon_n \to 0$ as $n \to \infty$.

Step 2: The tube

For every $x \in [a, b]$: $\boxed{f_n(x) - \varepsilon_n \le f(x) \le f_n(x) + \varepsilon_n.}$

Proof of Theorem 7.16

Step 3: Integrate the tube (monotonicity)

$$\int_a^b (f_n - \varepsilon_n)\, d\alpha \le \underline{\int_a^b} f\, d\alpha \le \overline{\int_a^b} f\, d\alpha \le \int_a^b (f_n + \varepsilon_n)\, d\alpha. \tag{25}$$

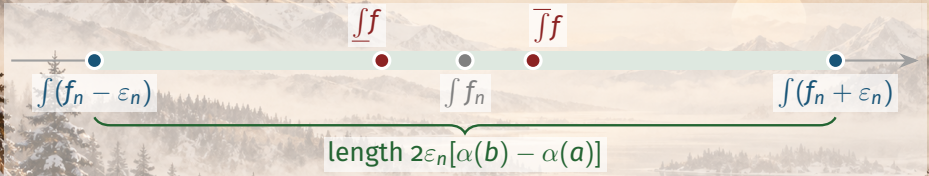
Outer inequalities: monotonicity. Middle inequality: Theorem 6.5.

Proof of Theorem 7.16

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Outer inequalities: monotonicity. Middle inequality: Theorem 6.5.



Proof of Theorem 7.16

Step 4: The gap closes, so $f \in \mathscr{R}(\alpha)$

Subtracting the outer ends of (25):

$$0 \le \overline{\int_a^b} f\,d\alpha - \underline{\int_a^b} f\,d\alpha \le 2\varepsilon_n[\alpha(b) - \alpha(a)] \xrightarrow[n\to\infty]{} 0.$$

The left side does not depend on n, so it is 0: $\boxed{f \in \mathscr{R}(\alpha).}$

Proof of Theorem 7.16

Step 5: The integrals converge

Now $\underline{\int} f = \overline{\int} f = \int_a^b f\,d\alpha$, and both $\int f\,d\alpha$ and $\int f_n\,d\alpha$ lie in the interval of (25):

$$\left| \int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha \right| \leq \varepsilon_n[\alpha(b) - \alpha(a)]. \tag{26}$$

Proof of Theorem 7.16

Step 5: The integrals converge

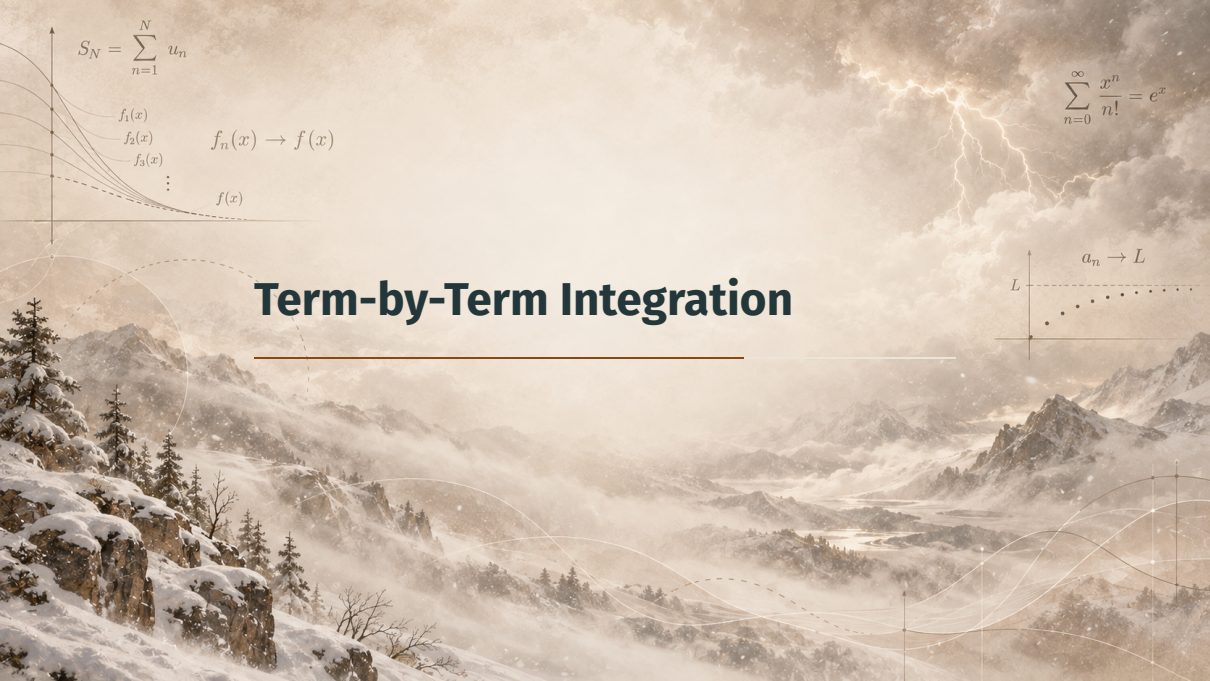
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$$\left| \int_a^b f\,d\alpha - \int_a^b f_n\,d\alpha \right| \leq \varepsilon_n[\alpha(b) - \alpha(a)]. \tag{26}$$

Since $\varepsilon_n \to 0$, this implies (23):

$$\boxed{\int_a^b f\,d\alpha = \lim_{n\to\infty} \int_a^b f_n\,d\alpha.}$$

■



Term-by-Term Integration

Corollary: Integrating Series Term by Term

Corollary (Theorem 7.16)

If $f_n \in \mathscr{R}(\alpha)$ on $[a, b]$ and if $f(x) = \sum_{n=1}^{\infty} f_n(x) \quad (a \le x \le b)$, the series converging uniformly on $[a, b]$, then

$$\int_a^b f \, d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n \, d\alpha.$$

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Why

Partial sums $s_N = \sum_{n=1}^{N} f_n \in \mathscr{R}(\alpha)$ (Theorem 6.12),
$\int s_N \, d\alpha = \sum_{n=1}^{N} \int f_n \, d\alpha$, $s_N \to f$ uniformly. Apply Theorem 7.16 to $\{s_N\}$.

Example: Integrating a Geometric Series

Example

Fix $0 < x < 1$. On $[0, x]$: $|t^n| \le x^n$ and $\sum x^n < \infty$, so

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n \quad \text{uniformly on } [0, x] \quad \text{(Theorem 7.10)}.$$

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Integrating term by term over $[0, x]$:

$$\int_0^x \frac{dt}{1-t} = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots$$

$f_n(x) \to f(x)$
$(n \to \infty)$



Remarks

$$S_N = \sum_{n=1}^{N} u_n$$



Revisiting Example 7.6

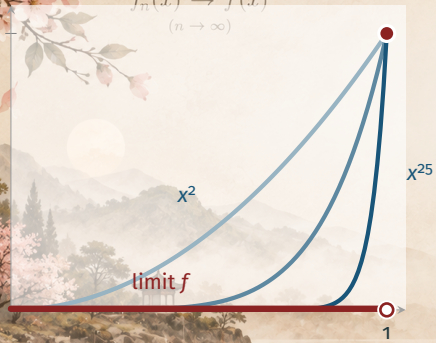
Remark: why Theorem 7.16 does not apply

For $f_n(x) = n\,x(1-x^2)^n$, the peak is at $x = 1/\sqrt{2n+1}$:

$$\sup_{0\le x\le 1} |f_n(x) - 0| = \frac{n}{\sqrt{2n+1}}\left(1 - \frac{1}{2n+1}\right)^n \approx \sqrt{\frac{n}{2e}} \to \infty.$$

So $f_n \to 0$ not uniformly. Conversely, Theorem 7.16 alone *proves* non-uniformity: $\int_0^1 f_n \to \frac{1}{2} \neq 0$.

Uniform Convergence Is Sufficient, Not Necessary



Remark: the converse fails

$f_n(x) = x^n$ on $[0, 1]$: the limit f is 0 on $[0, 1)$ and $f(1) = 1$.

- Not uniform (Theorem 7.12: f is discontinuous)
- Yet $\displaystyle\int_0^1 x^n \, dx = \frac{1}{n+1} \to 0 = \int_0^1 f \, dx$

Why the Bounded Interval Matters

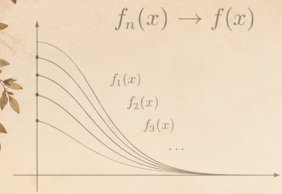


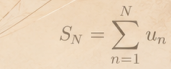
Remark: compact interval

On $[0, \infty)$ let $f_n = \frac{1}{n}$ on $[0, n]$, 0 elsewhere.

- $\sup |f_n| = \frac{1}{n} \to 0$ (uniform)
- but $\int_0^\infty f_n \, dx = 1$ for *every* n

(improper integrals; the interval has infinite length)







$\lim_{n\to\infty} u_n = L$

Summary



Summary

Key takeaways from this section:

- **Theorem 7.16:** $f_n \in \mathscr{R}(\alpha)$, $f_n \to f$ uniformly on $[a, b] \Rightarrow f \in \mathscr{R}(\alpha)$ and $\int f \, d\alpha = \lim \int f_n \, d\alpha$

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- **Proof:** tube $f_n \pm \varepsilon_n$ squeezes upper and lower integrals; error $\leq \varepsilon_n[\alpha(b) - \alpha(a)]$
- **Corollary:** uniformly conv. series can be integrated term by term

Coming up next: uniform convergence and differentiation.